

AUTOSAR Body Control Module (BCM) High-Level Design (Version 2.0 Revision)

[SAMPLE / TEST DATA FOR REVISION COMPARISON DEMONSTRATION ONLY] | Document Version: 2.0

1. System Purpose & Functional Scope

The Body Control Module (BCM) V2.0 manages vehicle interior lighting, door locks, and power window position control. Version 2.0 introduces automated power window anti-pinch algorithms and enhanced diagnostic logging.

2. Application Components

Components defined include **LightingControl_SWC**, **DoorLock_SWC**, and new component **WindowControl_SWC**. **LightingControl_SWC** receives headlight switch status over CAN. **WindowControl_SWC** regulates window motor position.

3. Interface & Port Configuration

DoorLock_SWC exposes **RPort_DoorStatus** and **PPort_LockActuator**. Interface **If_DoorState** conveys lock state via signal **Sig_DoorLock_status**. New component **WindowControl_SWC** exposes **PPort_WindowPos** over new interface **If_WindowPosition** with signal **Sig_WindowPos_pct**. Obsolete legacy interfaces and diagnostic signals from Version 1.0 have been removed in Version 2.0.

4. Basic Software Stack & Diagnostics

The BSW stack includes **EcuM**, **BswM**, **CanIf**, **Com**, and new module **Dem** (Diagnostic Event Manager).

5. Power Window Safety Subsystem

Section 5 is added in Version 2.0 to document anti-pinch current sensing and automated motor reverse logic.